

# SCAN Foundations

PSYCH 511.003 Spring 2025

2025-12-05

## Foundations of Social, Cognitive, and Affective Neuroscience (SCAN)

### Instructor

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### Where & When

467 Moore Thursdays, 1-4 PM

### About the course

The first scientific psychologists were physiologists fascinated by the possibility of understanding the mind by studying the brain. In this course, we will explore the historical roots and contemporary challenges associated with the study of biological approaches to complex adaptive behavior. In doing so, we will read and examine critically primary source readings that discuss basic patterns and processes of brain structure and function. The goal is to provide students with a basic foundation of knowledge about the structures and functions of the nervous system that can provide the basis for future study.

This course is one of two required courses for the [Specialization in Cognitive and Affective Neuroscience \(SCAN\)](#).

## Prerequisites

Undergraduate coursework in neuroscience or physiological psychology such as the equivalents of PSYCH 260 or BIO 469/470.

## Week 01: Jan 12-16

Thu, Jan 15

### Topic(s)

- Structure of the course
- Levels of analysis
- Causality in brain and behavior
- Does neuroscience need behavior? What does psychology need?

### Readings

- Required:
  - Krakauer, Ghazanfar, Gomez-Marin, MacIver, & Poeppel (2017)
  - Parada & Rossi (2018)
  - Newcombe (2017)
- Optional:
  - Siddiqi, Kording, Parvizi, & Fox (2022)
  - Churchland & Sejnowski (1988)
  - Favela (2020)
  - Ross & Bassett (2024)

### Materials

- [Slides](#)

### Assignments

- *Assigned:* [Exercise LVL](#). *Due:* 2026-01-22. [Canvas dropbox](#).

## **Week 02: Jan 19-23**

**Thu, Jan 22**

### **Topics**

- Neuroanatomy

### **Readings**

- [Neuroanatomy notes](#)

### **Materials**

- Slides. PDF
- [Allen Brain Atlas](#).

### **Assignments**

- *Due:* [Exercise LVL](#). Canvas dropbox
- *Assigned:* [Exercise ANAT](#). *Due:* 2026-01-29. [Canvas dropbox](#)

## **Week 03: Jan 26-30**

**Thu, Jan 29**

### **Topic(s)**

- Action

### **Reading(s)**

- Shenoy, Sahani, & Churchland (2013)

### **Materials**

- [Action notes](#)

### **Assignments**

- *Due:* [Exercise ANT](#). Canvas dropbox

## **Week 04: Feb 2-6**

**Thu, Feb 05**

### **Topic(s)**

- Perception

### **Readings**

- Khalsa et al. (2018)
- Shenoy et al. (2013)

### **Materials**

- [Perception notes](#)

### **Assignment(s)**

- *Assigned:* [Final project proposal](#). *Due:* 2026-03-05. [Canvas dropbox](#).
- *Assigned:* [Exercise PA](#). *Due:* 2026-02-12. [Canvas dropbox](#)

## **Week 05: Feb 9-13**

**Thu, Feb 12**

### **Topic(s)**

- Learning & memory

### **Readings**

### **Materials**

### **Assignment(s)**

- *Due:* [Exercise PA](#). [Canvas dropbox](#)

## **Week 06: Feb 16-20**

**Thu, Feb 19**

### **Topic(s)**

- Emotion

### **Readings**

- Malezieux, Klein, & Gogolla (2023)
- Watabe-Uchida, Eshel, & Uchida (2017)

### **Materials**

- [Notes](#)

### **Assignment(s)**

- *Assigned:* [Exercise EMO](#). *Due:* 2026-02-26. [Canvas dropbox](#)

## **Week 07: Feb 23-27**

**Thu, Feb 26**

### **Topic(s)**

- Sleep & circadian rhythms

### **Readings**

### **Materials**

### **Assignment(s)**

- *Due:* [Exercise EMO](#). [Canvas dropbox](#)
- *Assigned:* [Exercise ZZ](#). *Due:* 2026-03-05. [Canvas dropbox](#)

## Readings

## Materials

### Week 08: Mar 2-6

Thu, Mar 5

#### Topic(s)

- Evolution of the nervous system

## Readings

- Required:
  - Charvet & Finlay (2012)
  - Hofman (2014)
- Optional:
  - Castrillon et al. (2023)

## Materials

- [Notes](#)

#### Assignment(s)

- *Assigned:* [Exercise EVO](#). *Due:* 2026-03-19. [Canvas dropbox](#)
- *Due:* [Final project proposal](#). [Canvas dropbox](#)
- *Due:* [Exercise ZZ](#). [Canvas dropbox](#)

### Spring Break: Mar 9-13

NO CLASS

## Week 09: Mar 16-20

Thu, Mar 19

### Topic(s)

- Development of the nervous system

### Readings

- Required:
  - Cao, Huang, & He (2017)
  - Blumberg & Adolph (2023)
- Optional:
  - Larsen, Sydnor, Keller, Yeo, & Satterthwaite (2023)
  - Rakic (2009)

### Materials

- [Notes](#)

### Assignment(s)

- *Due:* [Exercise EVO](#). [Canvas dropbox](#).
- *Assigned:* [Exercise DEV](#). *Due:* 2026-03-26. [Canvas dropbox](#)

## Week 10: Mar 23-27

Thu, Mar 26

### Topics

- Cellular neuroscience I
  - Anatomy
  - Physiology
    - \* Resting potential

## Readings

- [Cellular neuroscience notes](#) | [PDF](#) |
- Optional:
  - Zeng & Sanes (2017)
  - Oliveira, Sardinha, Guerra-Gomes, Araque, & Sousa (2015)
  - Distéfano-Gagné, Bitarafan, Lacroix, & Gosselin (2023)

## Materials

- [Slides](#) | [PDF](#) |

## Assignment(s)

- *Due:* [Exercise DEV](#). [Canvas dropbox](#)
- *Assigned:* [Exercise PHYS](#). *Due:* 2026-04-02. [Canvas dropbox](#)

## Week 11: Mar 30 - Apr 3

### Thu, Apr 02

## Topics

- Cellular neuroscience II
  - Action potential
  - Synaptic transmission

## Readings

- [Cellular neuroscience notes](#)

## Materials

- Slides

## Assignment(s)

- *Due:* [Exercise PHYS](#). [Canvas dropbox](#)
- *Assigned:* [Exercise AP](#). *Due:* 2026-04-09. [Canvas dropbox](#)



## Week 12: Apr 6-10

Thu, Apr 09

### Topics

- Neurochemistry
  - Neurotransmitters
  - Hormones
- Neurocomputing

### Readings

- [Neurochemistry notes](#)
- Optional: Sarkar et al. (2016).

### Materials

- Slides

### Assignment(s)

- *Due:* [Exercise AP](#). [Canvas dropbox](#)
- *Assigned:* [Exercise CHEM](#). *Due:* 2026-04-16. [Canvas dropbox](#)

## Week 13: Apr 13-17

Thu, Apr 16

### Topics

- Disorder & disease I

### Readings

- Howes, Bukala, & Beck (2023)
- Volk, Chiu, Sharma, & Haganir (2015)

## Materials

- [Notes](#)

## Assignment(s)

- *Due:* [Exercise CHEM](#). [Canvas dropbox](#)

## Week 14: Apr 20-24

**Thu, Apr 23**

### Topics

- Disorder & disease II

### Readings

- Moncrieff et al. (2022)
- Namkung, Kim, & Sawa (2017)

## Materials

- [Notes](#)

## Week 15: Apr 27 - May 1

**Thu, Apr 30**

### Topics

- Frontiers in the biology of behavior
- Beethoven and the Cerebral Symphony

## Materials

- Slides

**Finals: May 4-8**

**Thu, May 07**

- [Final Project](#) write-up due | [Canvas dropbox](#)

## References

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